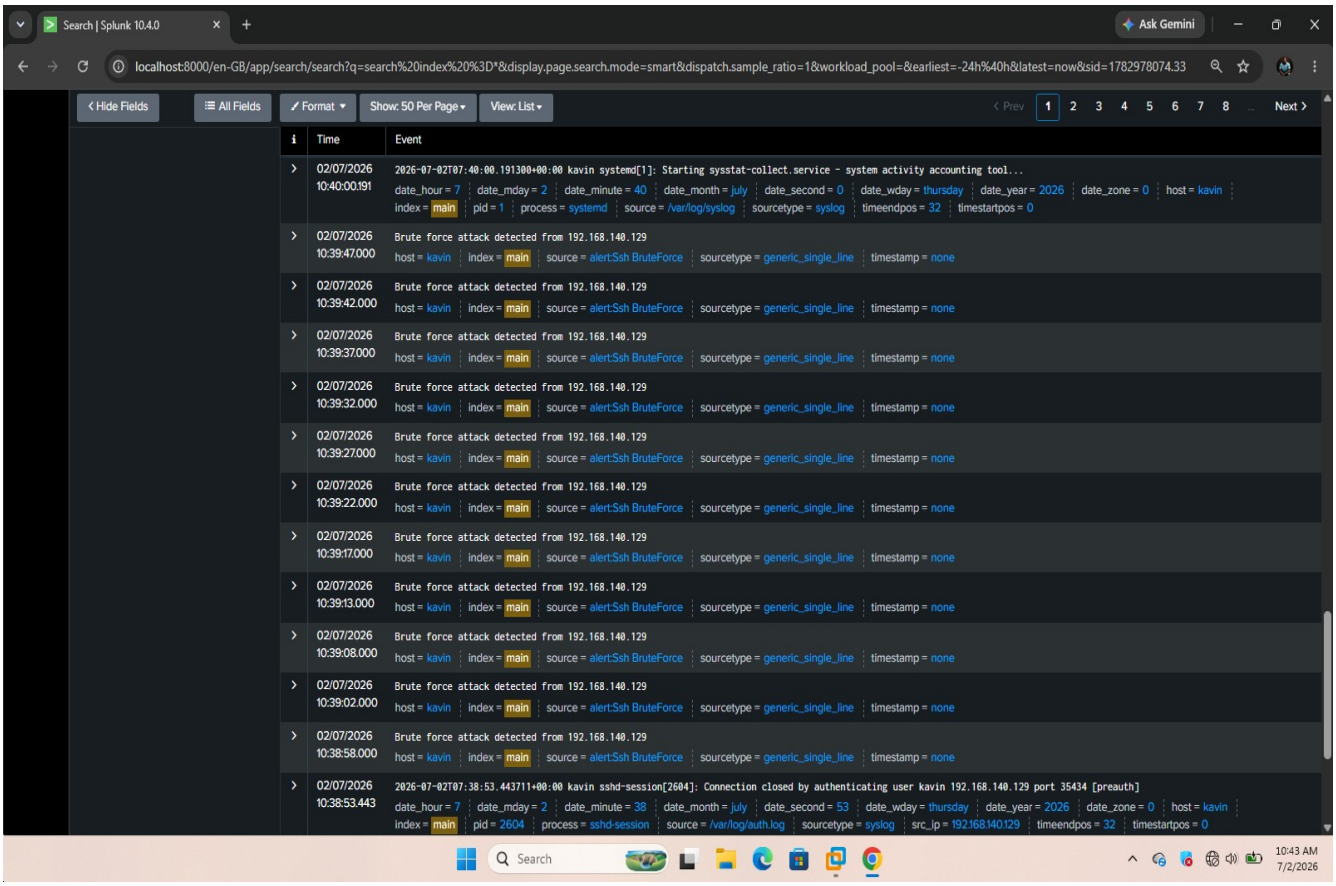
Using Splunk to Investigate a brute-force



Here we can see that we have a bruteforce attack that is happening on system or host named kavin.

Since I have already created an alert here we see that splunk already shows a presence of a bruce-force from the specif IP

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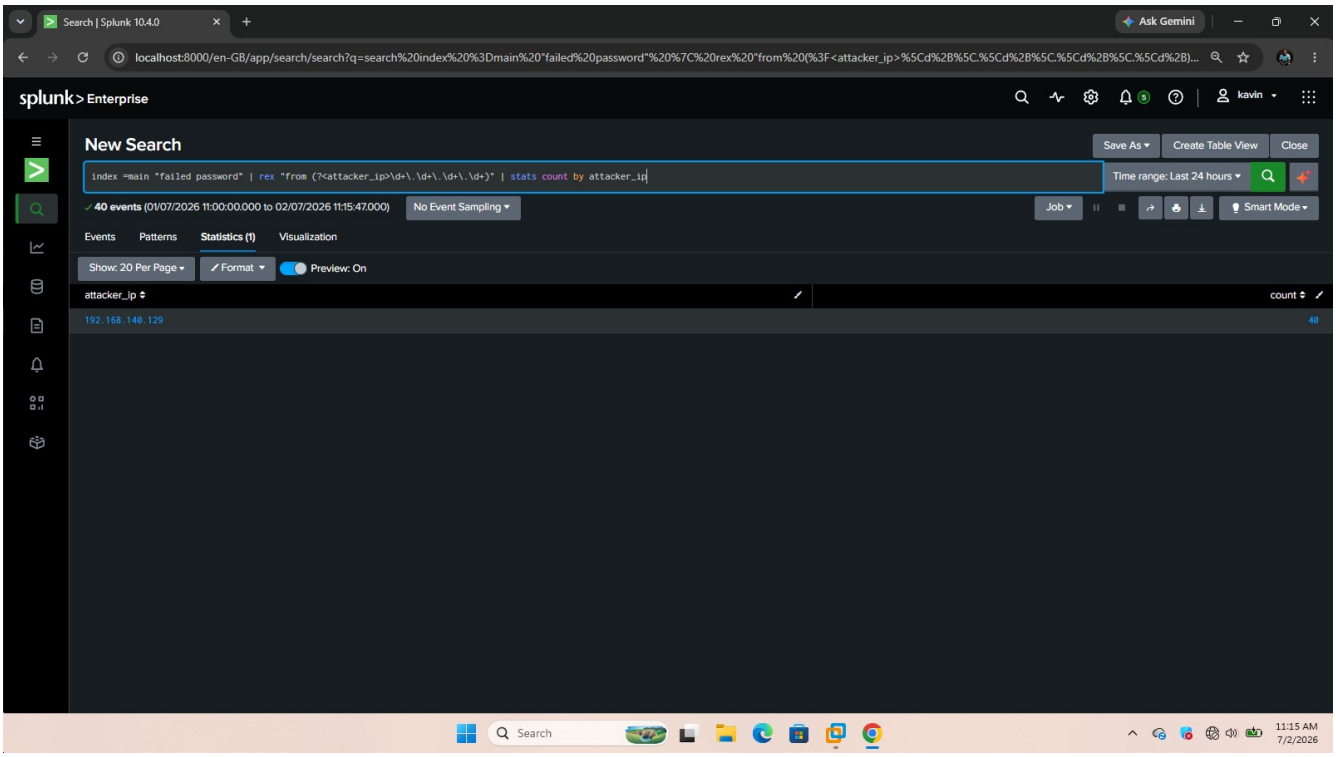

We used | *rex to extract fields from row logs.*
Good to know that /d means (0 – 9) so + means one more of the previous thing so \d+ =192 while \.=(.)dot hence it extracts our *ip* clearly

Then ?>attacker_ip> its used to match our ip with that name so that we can use field "attacker_ip" in our search and tables and even alerts later.

We see on the end that we have count which is displayed by command stats count by .
We have 40 attempts that could signal a compromised account.

Next question is was there any successful logins?lets go on and see also who does yhis account belong to

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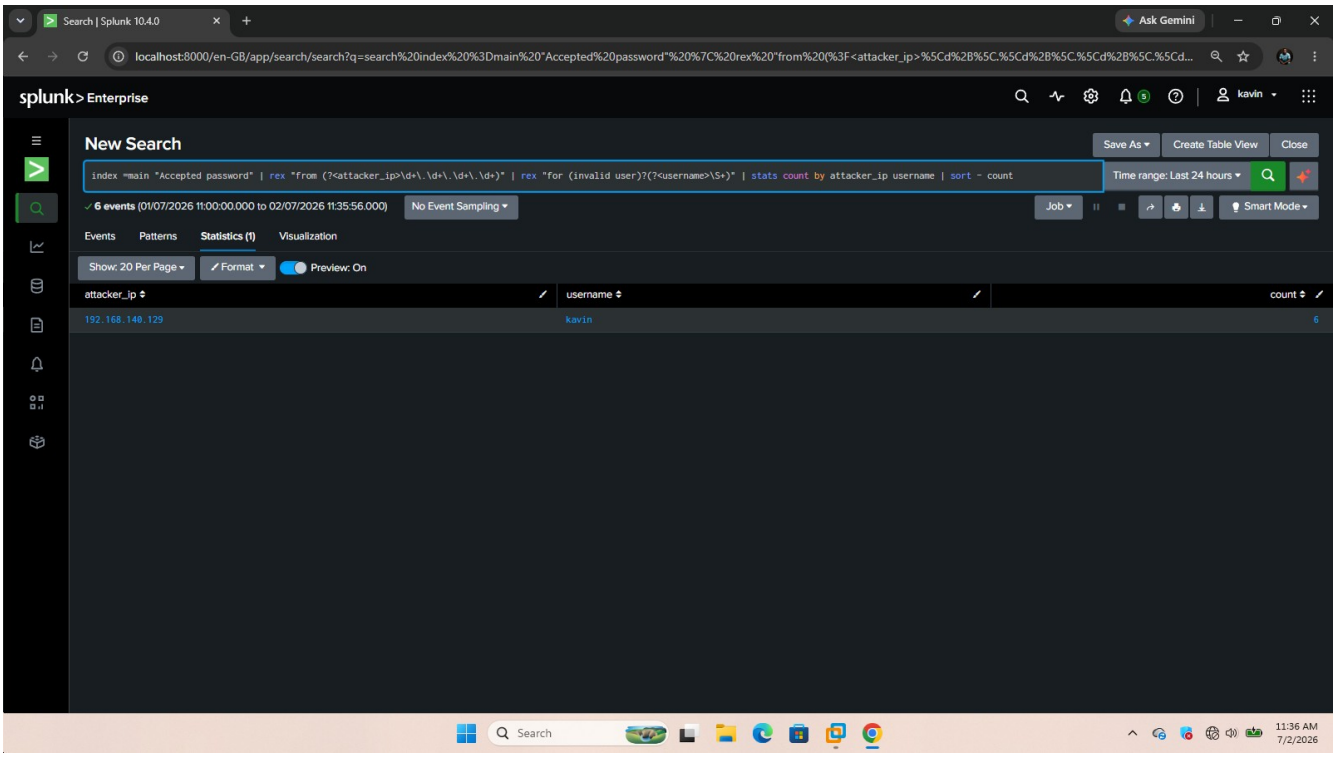

Now here is the smoking gun we have changed our search to accepted search shows that the account had 6 accepted logins.

Lets now see how our report would look like:

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Report Analysis:

Investigation	Results
Failed Logins	40
Successful Logins	6
Source IP	192.168.140.129
Username	kavin

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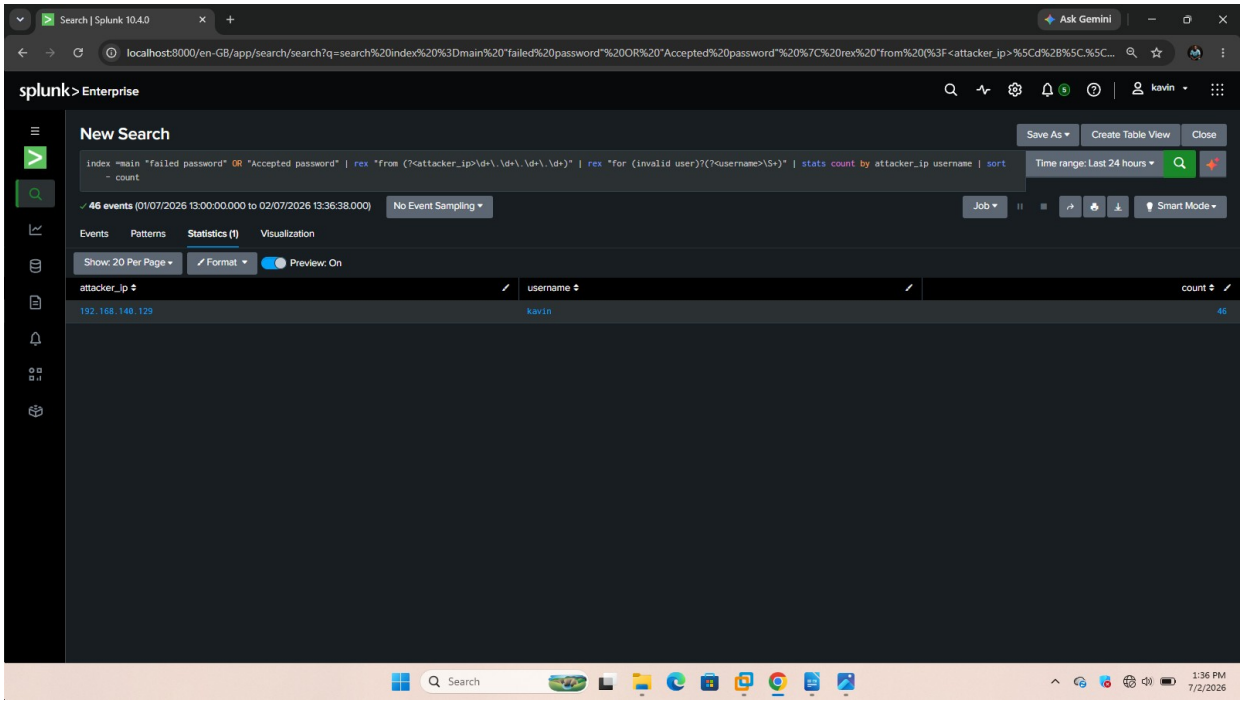

Here we build a Correlation Search:

And we see our total event is 46

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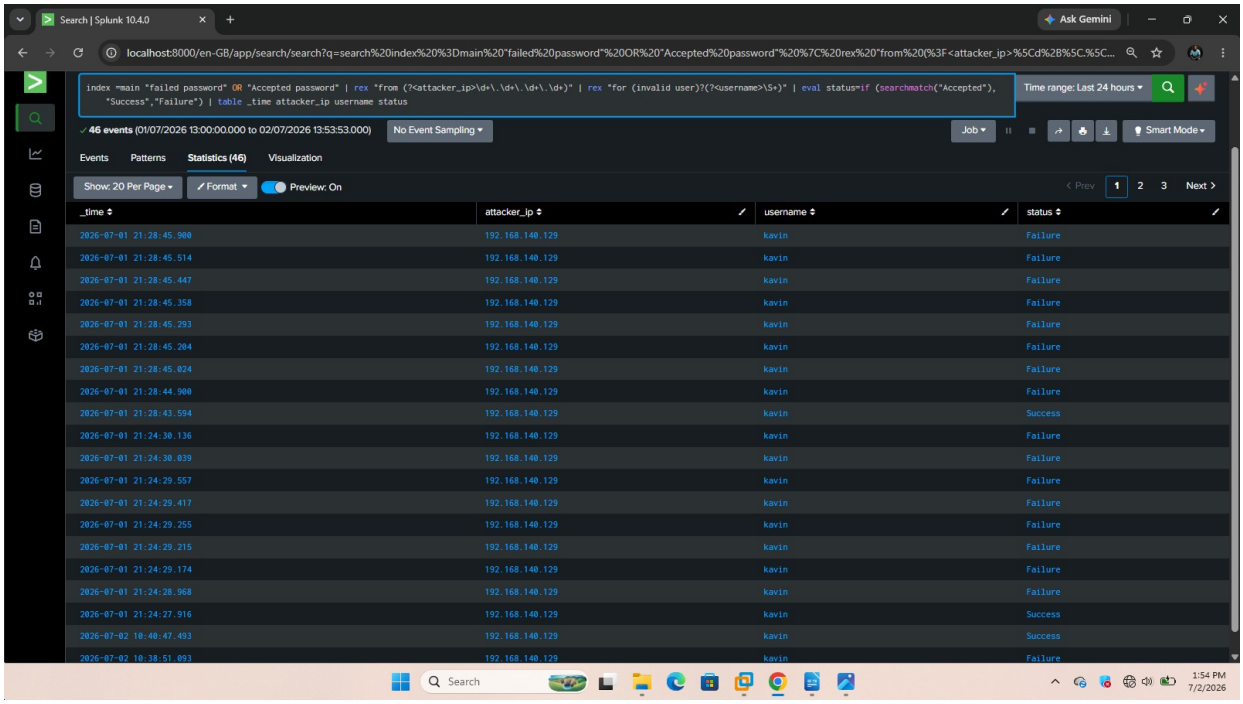


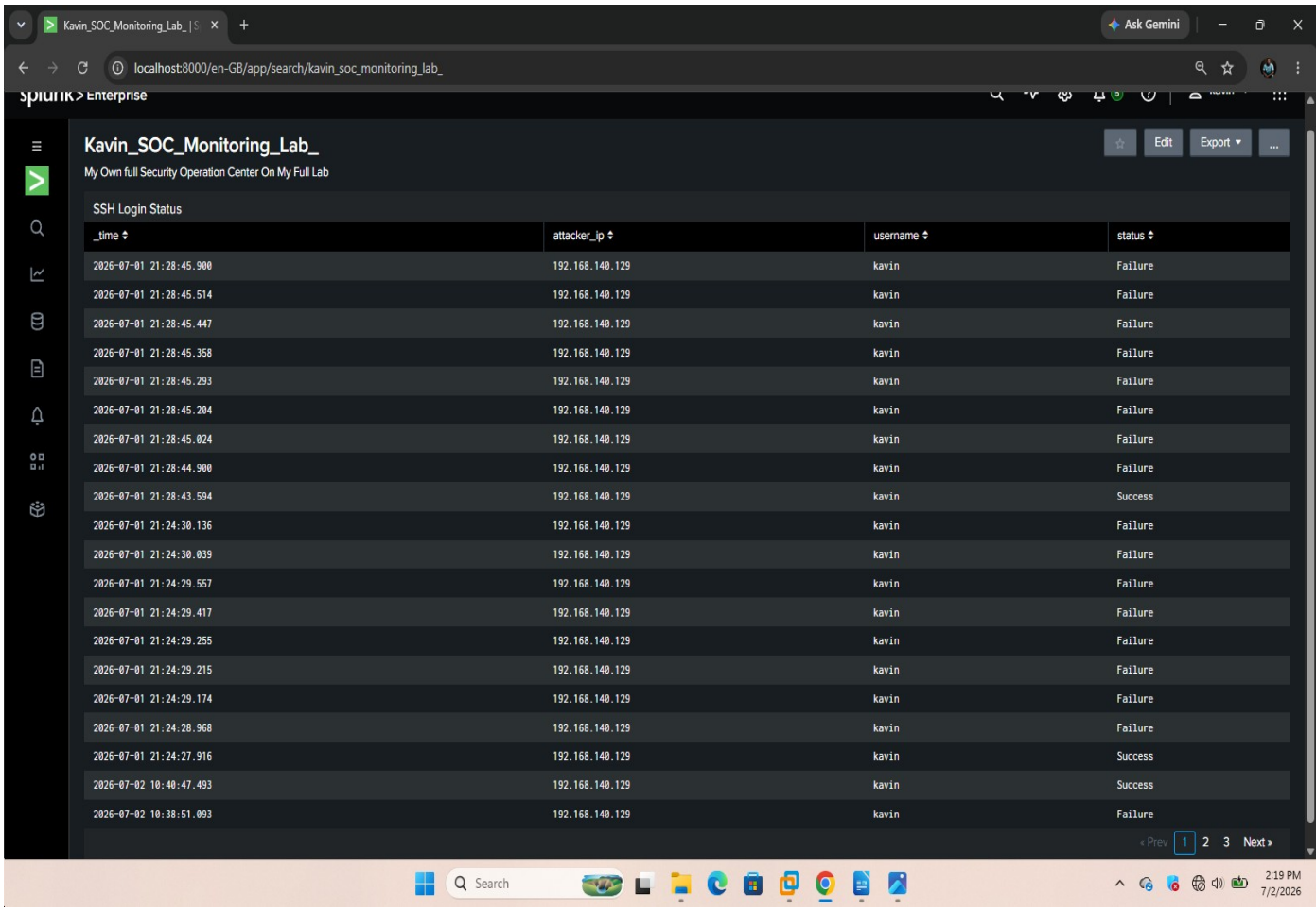
Now a new command eval :
Eval creates a new field like in our query *status=if (searchmatch("Accepted"), "Success","Failure")*

If the log contains *Accepted status is Successful*
Otherwise *status is failure*

_time **is for showing exact time**

Next I create a dashboard:

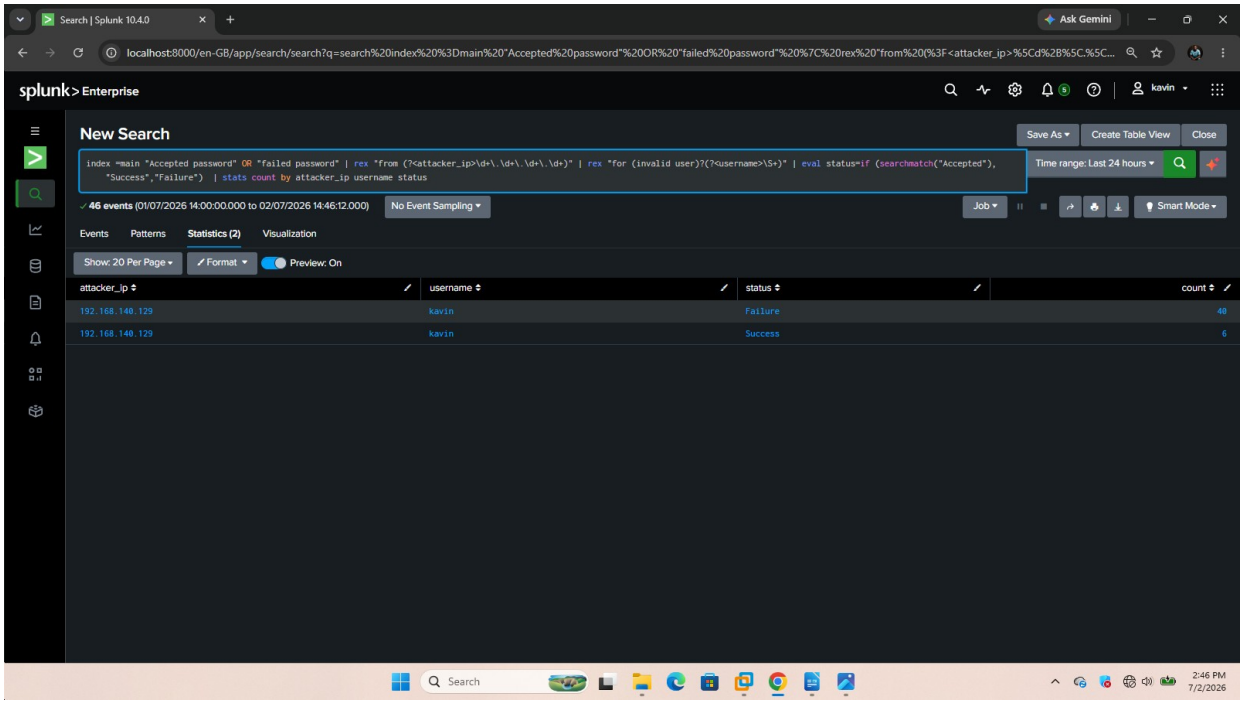
Using [Splunk] to Investigate a brute-force



Now here is my dashboard by the name Kavin_SOC_Monitoring_Lab.

With this we can add searches now.

Using Splunk to Investigate a brute-force



Now here we separated our successful logins with failed ones and we can see that all are coming from one IP.
We can see too that our search also counts each status as we added on search *status*

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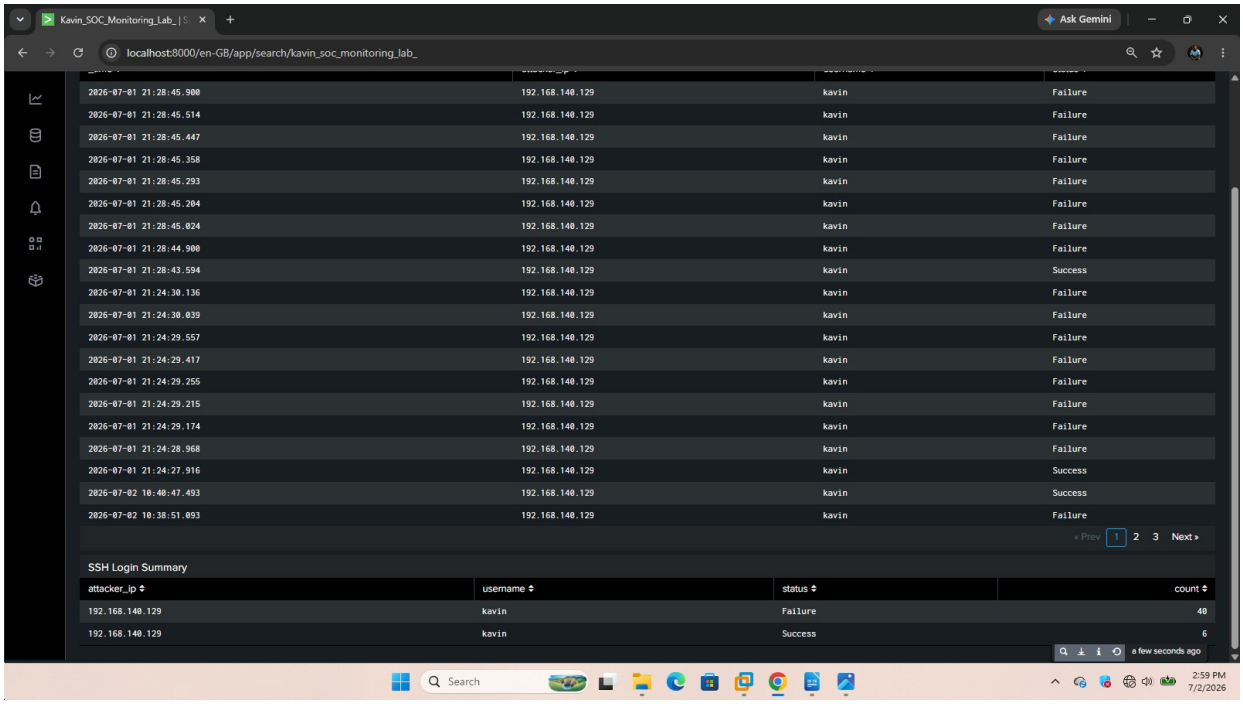
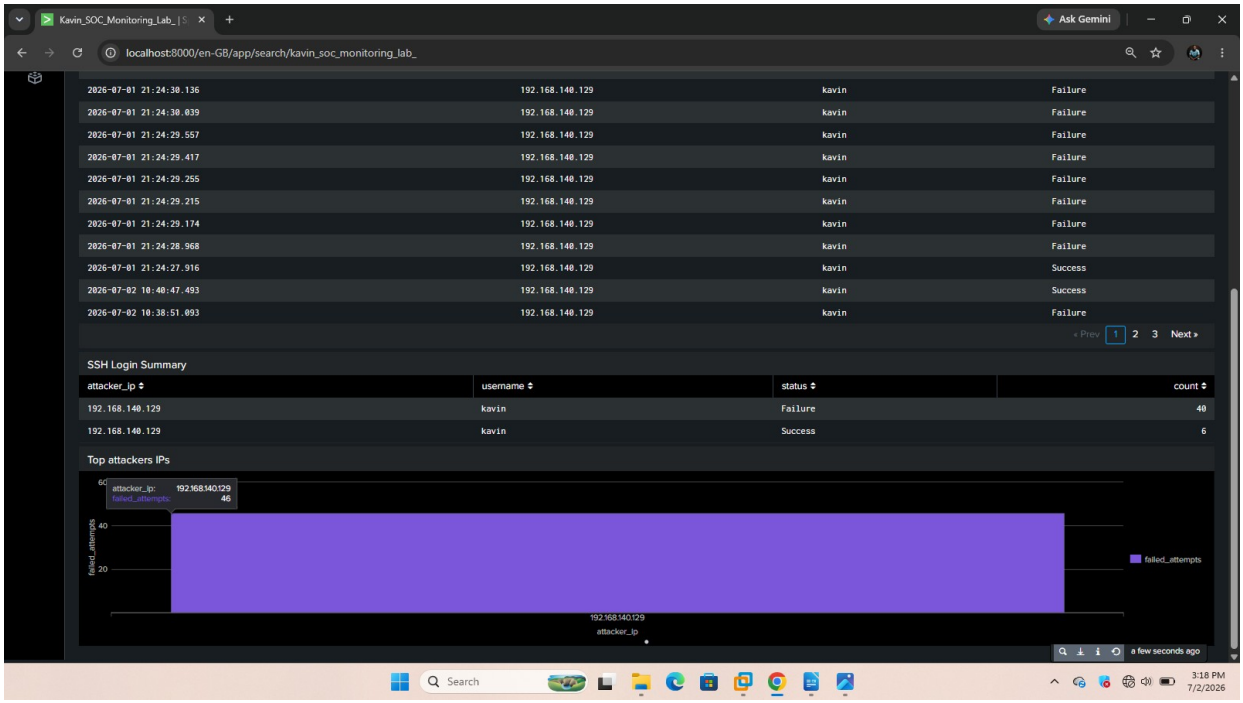


As we can see I have added a new a search on my existing dashboard and named it SSH summary, so that in future we can we can go back to it and we wont need to search same event we grab it and just use it.

Now its time I visualize my results:

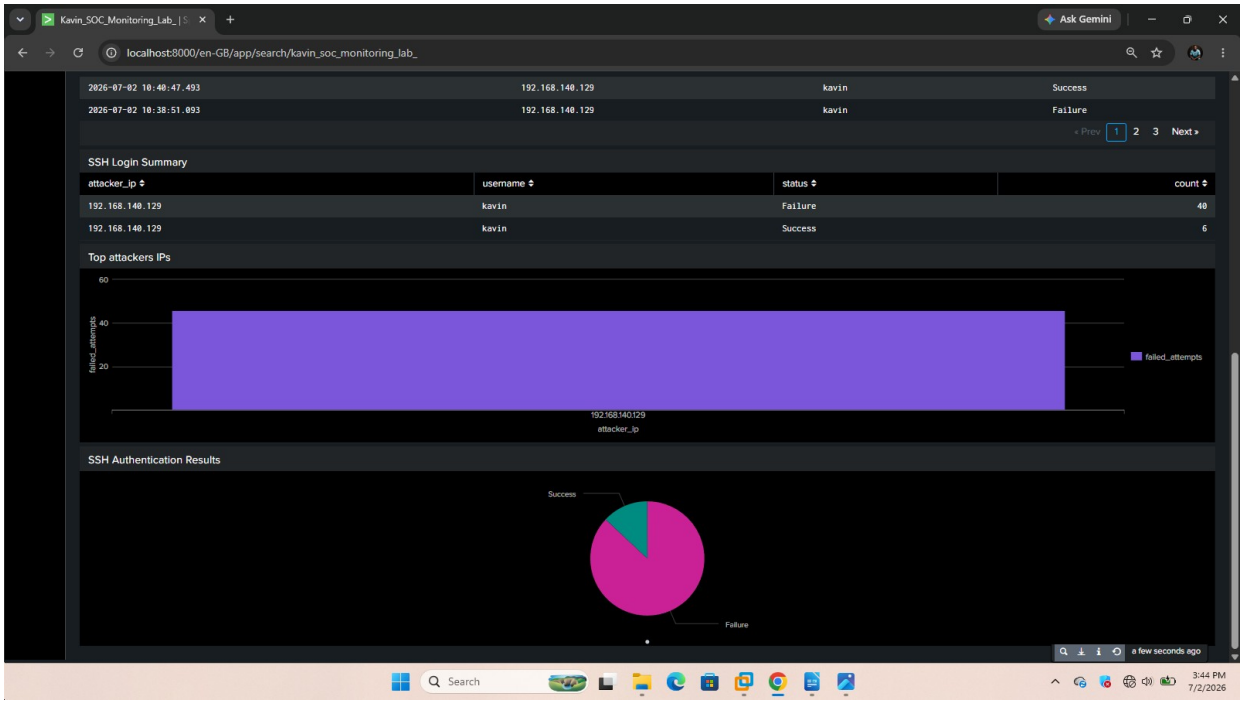
Using Splunk to Investigate a brute-force



Now we have added all our results as one bar to our existing dashboard

Now we separate things as real SOC:

Using Splunk to Investigate a brute-force



NOW as I look at the SSH Results as a Manager I can see that we had a lot of login attempts when I look at the *pie chart* but only few were successful.

Next is to investigate what made caused our system to break.

Using Splunk to Investigate a brute-force

SOC Monitoring Lab Report

Project Title: SSH Brute Force Detection and Monitoring using Splunk Enterprise

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Date: 2 July 2026

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1. Executive Summary

This project involved designing and implementing a Security Operations Center (SOC) monitoring solution using a home lab environment. The objective was to detect, investigate, and visualize SSH authentication activity on an Ubuntu Server using Splunk Enterprise.

The solution successfully collected system logs, identified failed and successful SSH authentication attempts, and presented security insights through a custom dashboard.

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2. Project Objective

The objectives of this project were:

- **Build a functional SOC monitoring environment.**
- **Collect Linux authentication logs.**
- **Detect SSH brute-force activity.**
- **Monitor successful and failed authentication events.**
- **Visualize security events using Splunk dashboards.**
- **Demonstrate SOC analyst investigation techniques.**

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3. Lab Environment

Component	Purpose
Kali Linux	Security testing workstation
Ubuntu Server	Target server (UF)

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Component	Purpose
UFW Firewall	Host firewall
Splunk Universal Forwarder	Log forwarding
Splunk Enterprise	SIEM platform

---

4. Network Architecture

```
Kali Linux
     |
SSH Authentication Attempts
     |
Ubuntu Server
     |
Authentication Logs
     |
Splunk Universal Forwarder
     |
Splunk Enterprise
     |
SOC Dashboard
```

---

5. Data Collection

Authentication logs generated on Ubuntu Server were forwarded to Splunk Enterprise using the Splunk Universal Forwarder.

The monitored log sources included Linux authentication events related to SSH login activity.

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6. Detection Engineering

Several Splunk Processing Language (SPL) searches were developed to support threat detection.

Examples included:

- **Detection of failed SSH logins**
- **Detection of successful SSH logins**
- **Extraction of attacker IP addresses**
- **Extraction of targeted usernames**
- **Classification of authentication events as Success or Failure**

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7. Investigation

The investigation focused on identifying authentication activity originating from the attacking host.

Findings

Source IP:

`192.168.140.129`

Target Username:

`kavin`

Failed Authentication Attempts:

`40`

Successful Authentication Attempts:

`6`

The successful authentication events were expected because they were generated during authorized testing within the lab environment.

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8. Dashboard Development

A custom Splunk dashboard named:

`Kavin_SOC_Monitoring_Lab`

was developed.

Dashboard panels included:

- **SSH Login Status**
- **SSH Login Summary**
- **SSH Authentication Results**

These panels provide both detailed event data and summarized authentication statistics.

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9. Skills Demonstrated

During this project the following cybersecurity skills were demonstrated:

- **Linux Administration**
- **SSH Security Monitoring**
- **Log Collection**
- **SIEM Operations**
- **Splunk Enterprise**
- **Splunk SPL**
- **Log Analysis**
- **Authentication Monitoring**
- **Threat Detection**
- **Security Dashboard Development**
- **Basic Threat Hunting**

---

10. Challenges Encountered

Several challenges were encountered during the project, including:

- **Understanding regular expressions** *(`rex`)*
- **Learning SPL aggregation using** `stats`
- **Choosing meaningful dashboard visualizations**
- **Distinguishing between detailed event tables and summarized charts**
- **Understanding how grouped fields affect search results**

These challenges were resolved through iterative testing and validation.

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11. Lessons Learned

This project demonstrated that effective SOC dashboards should communicate security information quickly and clearly.

Rather than displaying raw logs alone, dashboards should answer key operational questions such as:

- **Are authentication failures increasing?**
- **Which systems are under attack?**
- **Which accounts are being targeted?**

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- **Are authentication attempts succeeding?**

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12. Future Improvements

Future enhancements include:

- **Failed Login Trend Analysis**
- **Top Attacker IP Dashboard**
- **Top Targeted Usernames**
- **UFW Firewall Monitoring**
- **Automated Brute Force Alerts**
- **Email Notifications**
- **Automated Incident Response**
- **Threat Intelligence Integration**

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13. Conclusion

This project successfully demonstrated the deployment of a basic Security Operations Center monitoring solution using Splunk Enterprise. Authentication logs from an Ubuntu Server were collected, analyzed, and visualized to support detection of SSH authentication activity.

The project provided practical experience in SIEM configuration, log analysis, dashboard development, and basic threat detection, while establishing a foundation for more advanced SOC monitoring capabilities